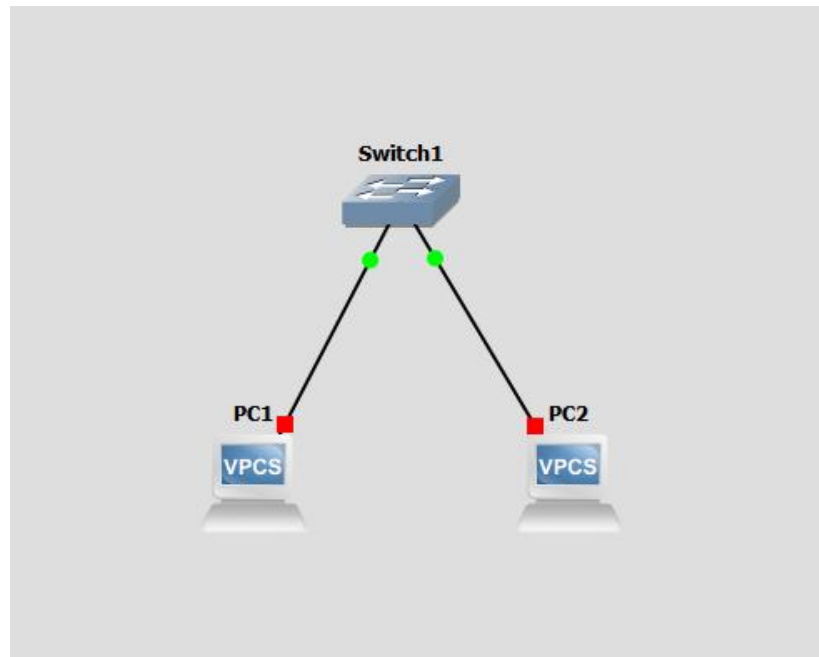


Лабораторная работа №1

Тема: Освоение инструментария для выполнения работ, построение простой сети

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети.



Для PC1: ip 192.168.1.1 255.255.255.0

Для PC2: ip 192.168.1.2 255.255.255.0

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера.

```
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

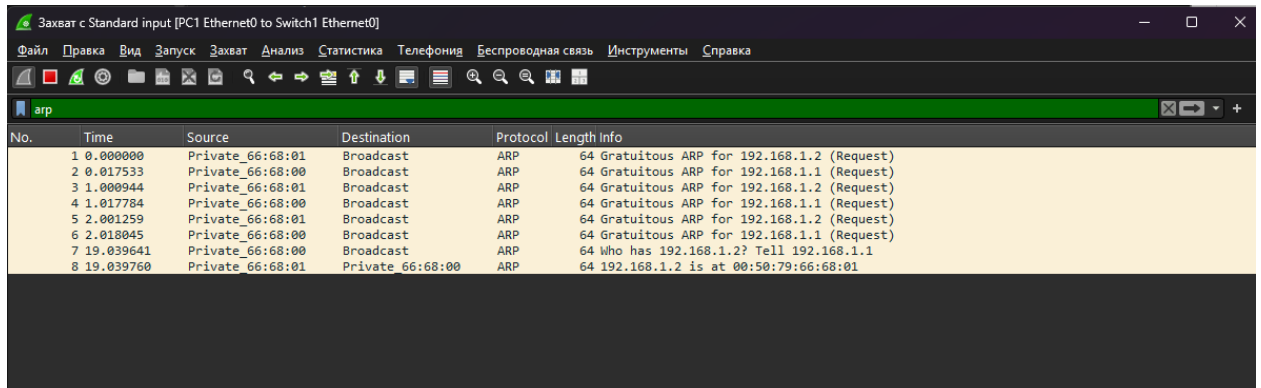
PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.220 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.251 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.226 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.242 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.235 ms

PC1> 
```

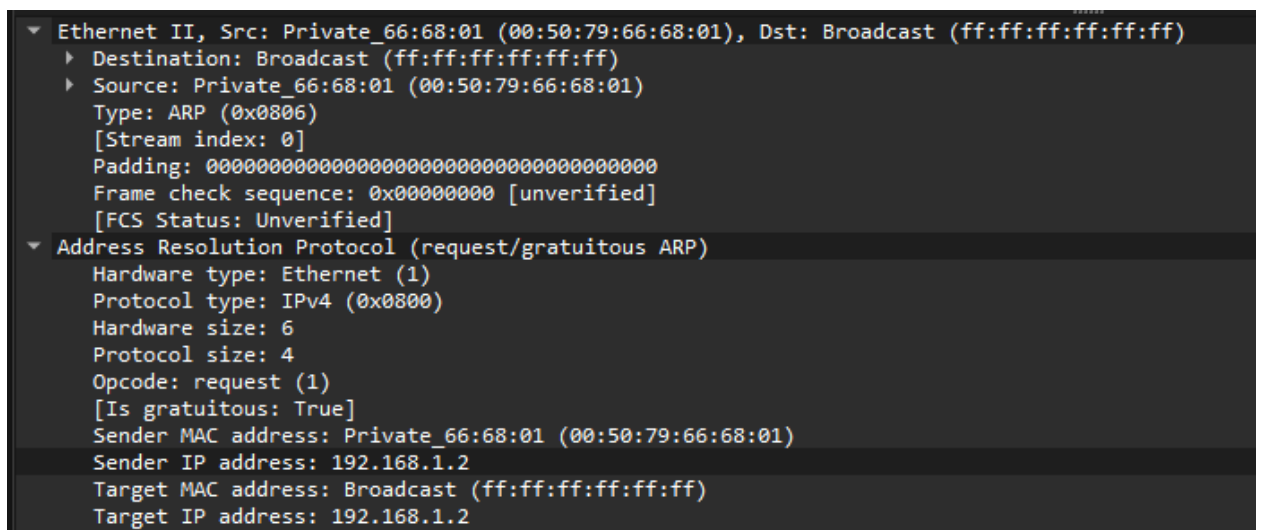
4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark.

Трафик на обоих линках одинаковый:



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.2 (Request)
2	0.017533	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.1 (Request)
3	1.000944	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.2 (Request)
4	1.017784	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.1 (Request)
5	2.001259	Private_66:68:01	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.2 (Request)
6	2.018045	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 192.168.1.1 (Request)
7	19.039641	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.2? Tell 192.168.1.1
8	19.039760	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.2 is at 00:50:79:66:68:01

первые 6 ARP-пакетов (Gratuitous ARP), нужны для предотвращения конфликтов ip адресов в сети: если придет ответ, значит адрес занят.



```
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ► Destination: Broadcast (ff:ff:ff:ff:ff:ff)
  ► Source: Private_66:68:01 (00:50:79:66:68:01)
    Type: ARP (0x0806)
    [Stream index: 0]
    Padding: 000000000000000000000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
  ▼ Address Resolution Protocol (request/gratuitous ARP)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    [Is gratuitous: True]
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 192.168.1.2
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.1.2
```

широковещательный запрос (Dest: Broadcast ff:ff:ff:ff:ff:ff).

Sender ip и target ip совпадают, потому что устройству необходимо получить ответ от другого узла, который использует такой же ip.

Who has 192.168.1.2? Tell 192.168.1.1

```

▶ Frame 7: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  ▼ Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... 1. .... = LG bit: Locally administered address (this is NOT the factory default)
    .... 1. .... = IG bit: Group address (multicast/broadcast)
  ▼ Source: Private_66:68:00 (00:50:79:66:68:00)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
  Padding: 00000000000000000000000000000000
  Frame check sequence: 0x00000000 [unverified]
  [FCS Status: Unverified]
▼ Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.1
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.2

```

Также широковещательный запрос (Dest: Broadcast ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.2 – ip устройства MAC адрес которого нужно узнать.

ARP ответ:

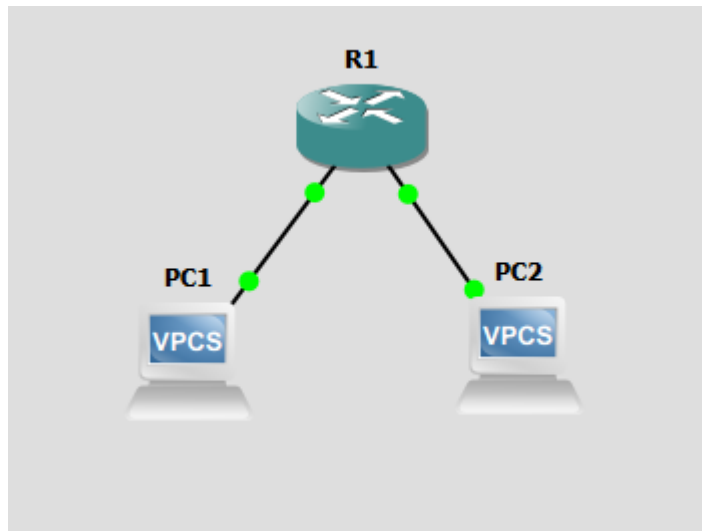
```

▶ Frame 8: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
  ▼ Destination: Private_66:68:00 (00:50:79:66:68:00)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  ▼ Source: Private_66:68:01 (00:50:79:66:68:01)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 2]
  Padding: 00000000000000000000000000000000
  Frame check sequence: 0x00000000 [unverified]
  [FCS Status: Unverified]
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.1.2
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.1.1

```

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01 – искомый MAC-адрес)

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей.



PC1: ip 192.168.1.10 255.255.255.0 192.168.1.1

PC2: ip 10.0.0.10 255.255.255.0 10.0.0.1

R1:

```
R1#
R1#Q
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:03:34.371: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:35.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#interface fastEthernet 1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar 1 00:03:59.635: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:04:00.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#end
R1#
*Mar 1 00:04:19.515: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

enable

configure terminal

interface fastEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface fastEthernet 1/0

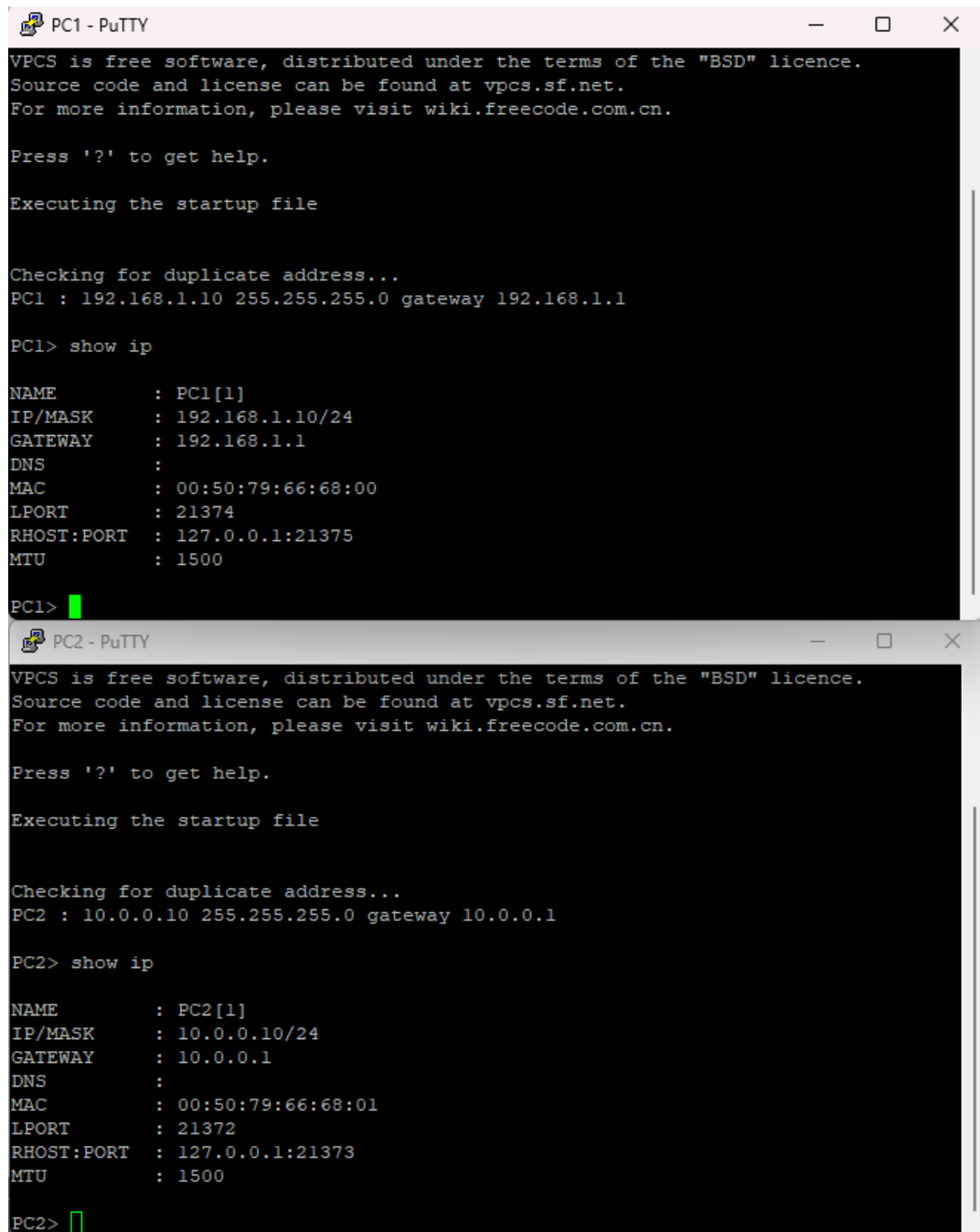
ip address 10.0.0.1 255.255.255.0

no shutdown

exit

end

write memory



The image shows two terminal windows, PC1 - PuTTY and PC2 - PuTTY, displaying the VPCS startup sequence and configuration. Both windows show the same initial text: 'VPCS is free software, distributed under the terms of the "BSD" licence. Source code and license can be found at vpcs.sf.net. For more information, please visit wiki.freecode.com.cn. Press '?' to get help. Executing the startup file'. PC1 then shows 'Checking for duplicate address...' followed by its configuration: 'PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1'. After the 'show ip' command, it displays network details for PC1[1], including IP/MASK (192.168.1.10/24), GATEWAY (192.168.1.1), DNS, MAC (00:50:79:66:68:00), LPORT (21374), RHOST:PORT (127.0.0.1:21375), and MTU (1500). PC2 follows a similar pattern with configuration: 'PC2 : 10.0.0.10 255.255.255.0 gateway 10.0.0.1' and network details for PC2[1], including IP/MASK (10.0.0.10/24), GATEWAY (10.0.0.1), DNS, MAC (00:50:79:66:68:01), LPORT (21372), RHOST:PORT (127.0.0.1:21373), and MTU (1500).

```
PC1 - PuTTY
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 21374
RHOST:PORT : 127.0.0.1:21375
MTU        : 1500

PC1>

PC2 - PuTTY
VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
PC2 : 10.0.0.10 255.255.255.0 gateway 10.0.0.1

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.0.0.10/24
GATEWAY    : 10.0.0.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 21372
RHOST:PORT : 127.0.0.1:21373
MTU        : 1500

PC2>
```

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера.

```

PC1> ping 10.0.0.10

10.0.0.10 icmp_seq=1 timeout
84 bytes from 10.0.0.10 icmp_seq=2 ttl=63 time=20.584 ms
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=15.134 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=15.025 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=15.499 ms

PC1> █

```

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark.

```

PC1> ping 10.0.0.10

84 bytes from 10.0.0.10 icmp_seq=1 ttl=63 time=30.028 ms
84 bytes from 10.0.0.10 icmp_seq=2 ttl=63 time=15.866 ms
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=15.229 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=15.211 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=15.321 ms

PC1> █

```

*Standard input [PC1 Ethernet0 to R1 FastEthernet0/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp || icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	8.374521	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
4	8.380669	cc:01:3b:72:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at cc:01:3b:72:00:00
5	8.380886	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdd35, seq=1/256, ttl=64 (reply in 6)
6	8.410836	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdd35, seq=1/256, ttl=63 (request in 5)
7	9.411428	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xde35, seq=2/512, ttl=64 (reply in 8)
8	9.427207	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xde35, seq=2/512, ttl=63 (request in 7)
10	10.428379	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdf35, seq=3/768, ttl=64 (reply in 11)
11	10.443525	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdf35, seq=3/768, ttl=63 (request in 10)
12	11.444411	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe035, seq=4/1024, ttl=64 (reply in 13)
13	11.459514	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe035, seq=4/1024, ttl=63 (request in 12)
14	12.460310	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe135, seq=5/1280, ttl=64 (reply in 15)
15	12.475541	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe135, seq=5/1280, ttl=63 (request in 14)

*Standard input [PC2 Ethernet0 to R1 FastEthernet1/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp || icmp

No.	Time	Source	Destination	Protocol	Length	Info
2	5.514111	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdd35, seq=1/256, ttl=63 (reply in 5)
3	5.514211	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.10
4	5.524167	cc:01:3b:72:00:10	Private_66:68:01	ARP	60	10.0.0.1 is at cc:01:3b:72:00:10
5	5.524810	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdd35, seq=1/256, ttl=64 (request in 2)
6	6.540494	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xde35, seq=2/512, ttl=63 (reply in 7)
7	6.540577	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xde35, seq=2/512, ttl=64 (request in 6)
9	7.556823	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdf35, seq=3/768, ttl=63 (reply in 10)
10	7.556933	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdf35, seq=3/768, ttl=64 (request in 9)
11	8.572842	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe035, seq=4/1024, ttl=63 (reply in 12)
12	8.572925	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe035, seq=4/1024, ttl=64 (request in 11)
13	9.588865	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe135, seq=5/1280, ttl=63 (reply in 14)
14	9.588965	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe135, seq=5/1280, ttl=64 (request in 13)

ARP протокол:

Заголовки аналогичны пункту 4, но узнаём MAC-адрес шлюза.

ICMP Запрос от PC1 до R1:

```
Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -,
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:3b:72:00:00 (cc:01:3b:72:00:00)
  Destination: cc:01:3b:72:00:00 (cc:01:3b:72:00:00)
  Source: Private_66:68:00 (00:50:79:66:68:00)
  Type: IPv4 (0x0800)
  [Stream index: 3]
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 10.0.0.10
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x35dd (13789)
  000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: ICMP (1)
  Header Checksum: 0x7910 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.10
  Destination Address: 10.0.0.10
  [Stream index: 0]
Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0x42d5 [correct]
  [Checksum Status: Good]
  Identifier (BE): 56629 (0xdd35)
  Identifier (LE): 13789 (0x35dd)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 6]
  Data (56 bytes)
```

ICMP инкапсулирован в IP.

Source: Private_66:68:00 (00:50:79:66:68:00) -MAC-адрес PC1

Destination: cc:01:3b:72:00:00 (cc:01:3b:72:00:00) - MAC адрес R1(маршрутизатор).

Source Address: 192.168.1.10 - ip PC1

Destination Address: 10.0.0.10 - ip PC2

Time to Live: 64

От R1 до PC2:

```

Frame 2: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -,
Ethernet II, Src: cc:01:3b:72:00:10 (cc:01:3b:72:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Destination: Private_66:68:01 (00:50:79:66:68:01)
  Source: cc:01:3b:72:00:10 (cc:01:3b:72:00:10)
  Type: IPv4 (0x0800)
  [Stream index: 1]
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 10.0.0.10
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x35dd (13789)
  000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0x7a10 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.10
  Destination Address: 10.0.0.10
  [Stream index: 0]
Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0x42d5 [correct]
  [Checksum Status: Good]
  Identifier (BE): 56629 (0xdd35)
  Identifier (LE): 13789 (0x35dd)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 5]
  Data (56 bytes)

```

Destination: Private_66:68:01 (00:50:79:66:68:01) -MAC-адрес PC2

Source: cc:01:3b:72:00:10 (cc:01:3b:72:00:10) - MAC адрес R1(маршрутизатор).

Source Address: 192.168.1.10 - ip PC1

Destination Address: 10.0.0.10 - ip PC2

Time to Live: 63 – снизился при прохождении маршрутизатора.